

Python Programs with Output Screenshots

Question 1

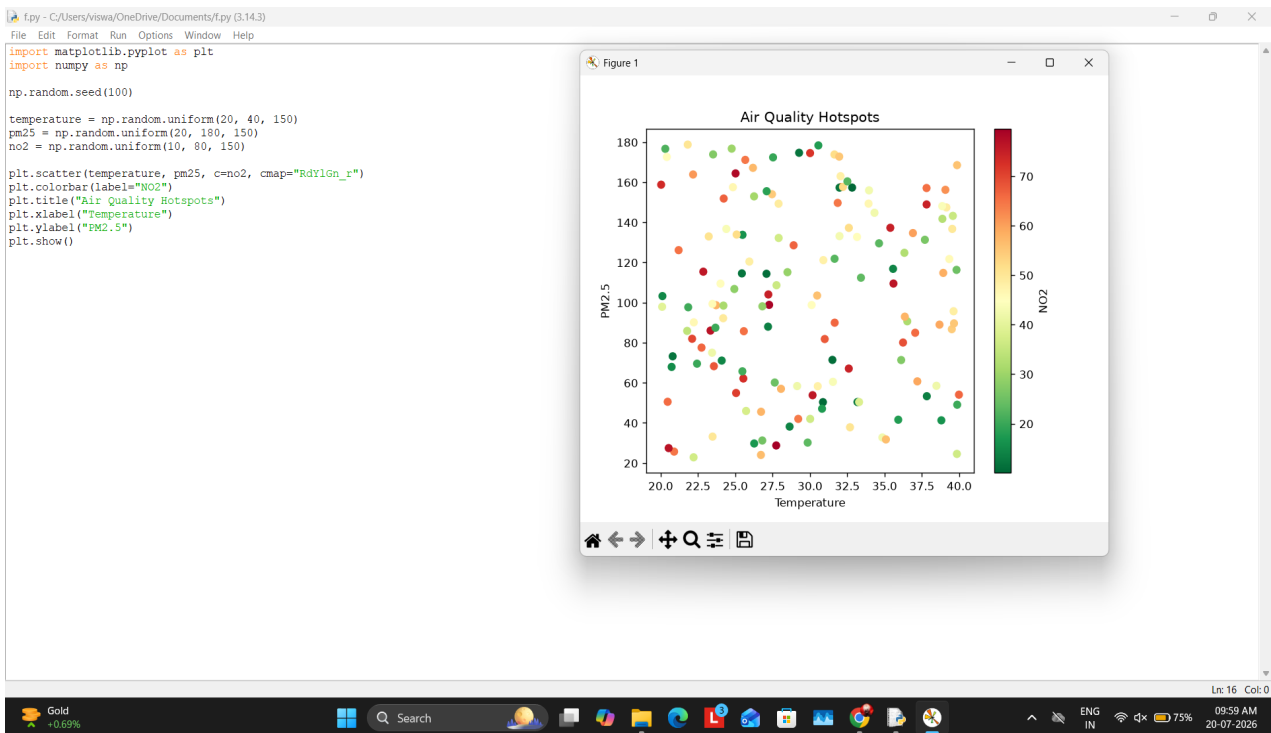
```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(100)

temperature = np.random.uniform(20, 40, 150)
pm25 = np.random.uniform(20, 180, 150)
no2 = np.random.uniform(10, 80, 150)

plt.scatter(temperature, pm25, c=no2, cmap="RdYlGn_r")
plt.colorbar(label="NO2")
plt.title("Air Quality Hotspots")
plt.xlabel("Temperature")
plt.ylabel("PM2.5")
plt.show()
```

Output Screenshot



Question 2

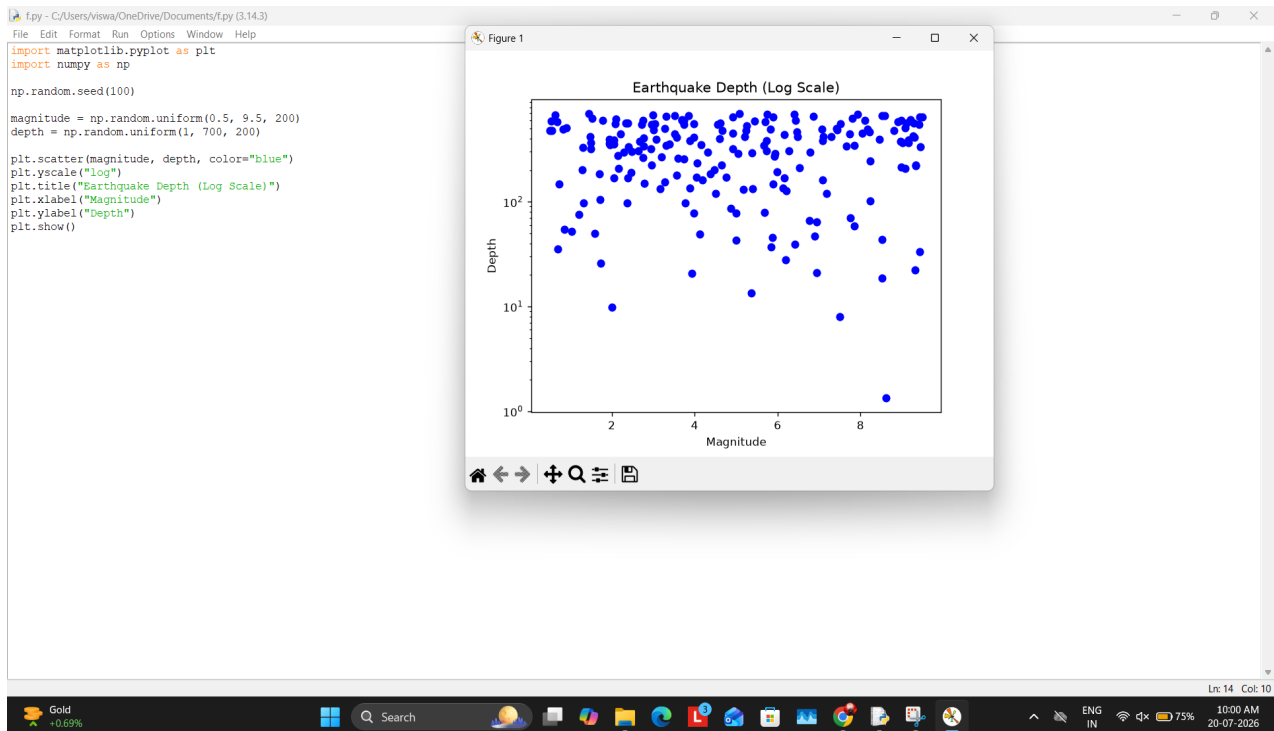
```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(100)

magnitude = np.random.uniform(0.5, 9.5, 200)
depth = np.random.uniform(1, 700, 200)

plt.scatter(magnitude, depth, color="blue")
plt.yscale("log")
plt.title("Earthquake Depth (Log Scale)")
plt.xlabel("Magnitude")
plt.ylabel("Depth")
plt.show()
```

Output Screenshot



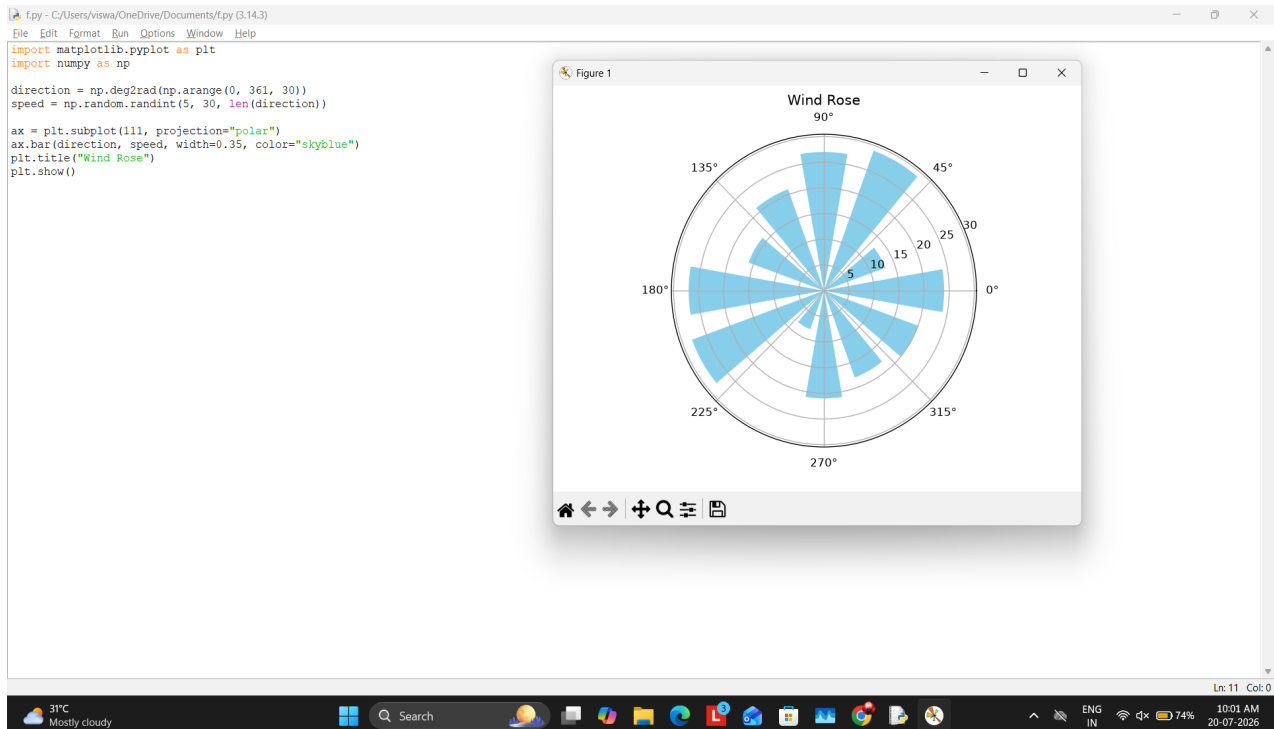
Question 3

```
import matplotlib.pyplot as plt
import numpy as np

direction = np.deg2rad(np.arange(0, 361, 30))
speed = np.random.randint(5, 30, len(direction))

ax = plt.subplot(111, projection="polar")
ax.bar(direction, speed, width=0.35, color="skyblue")
plt.title("Wind Rose")
plt.show()
```

Output Screenshot



Question 4

```
import matplotlib.pyplot as plt
import numpy as np

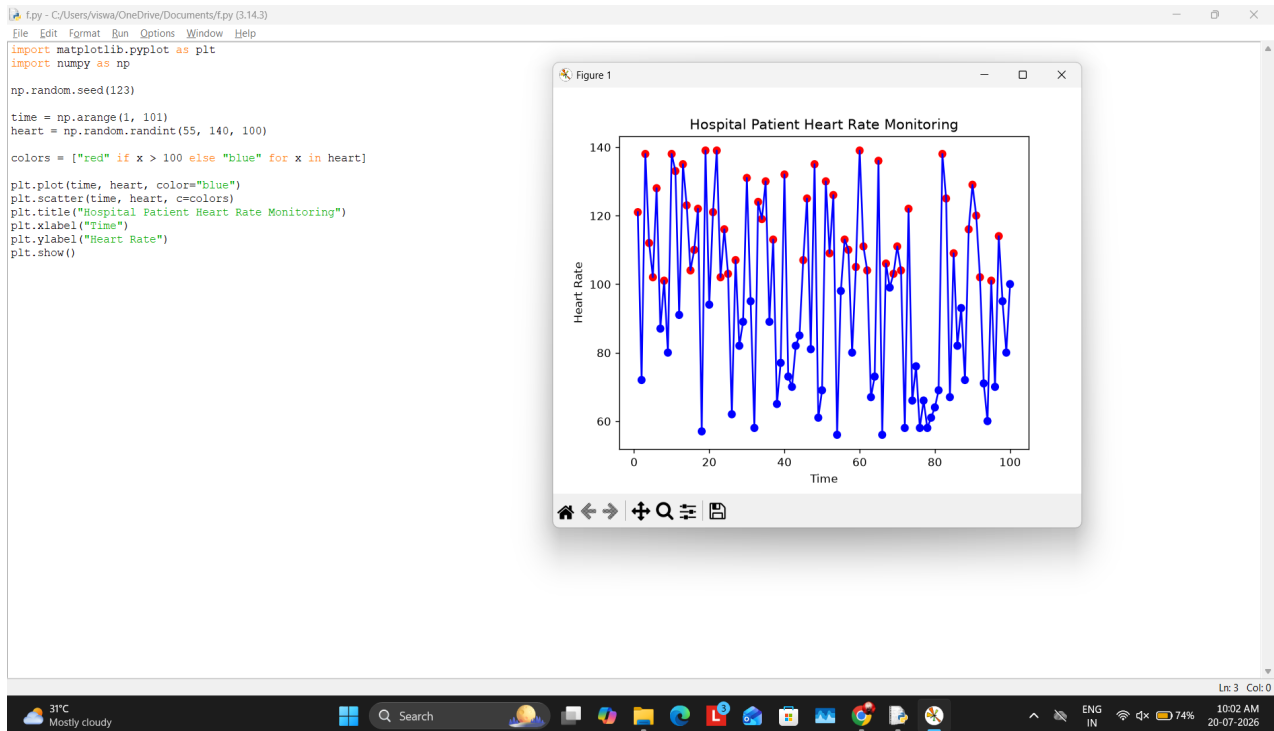
np.random.seed(123)

time = np.arange(1, 101)
heart = np.random.randint(55, 140, 100)

colors = ["red" if x > 100 else "blue" for x in heart]

plt.plot(time, heart, color="blue")
plt.scatter(time, heart, c=colors)
plt.title("Hospital Patient Heart Rate Monitoring")
plt.xlabel("Time")
plt.ylabel("Heart Rate")
plt.show()
```

Output Screenshot



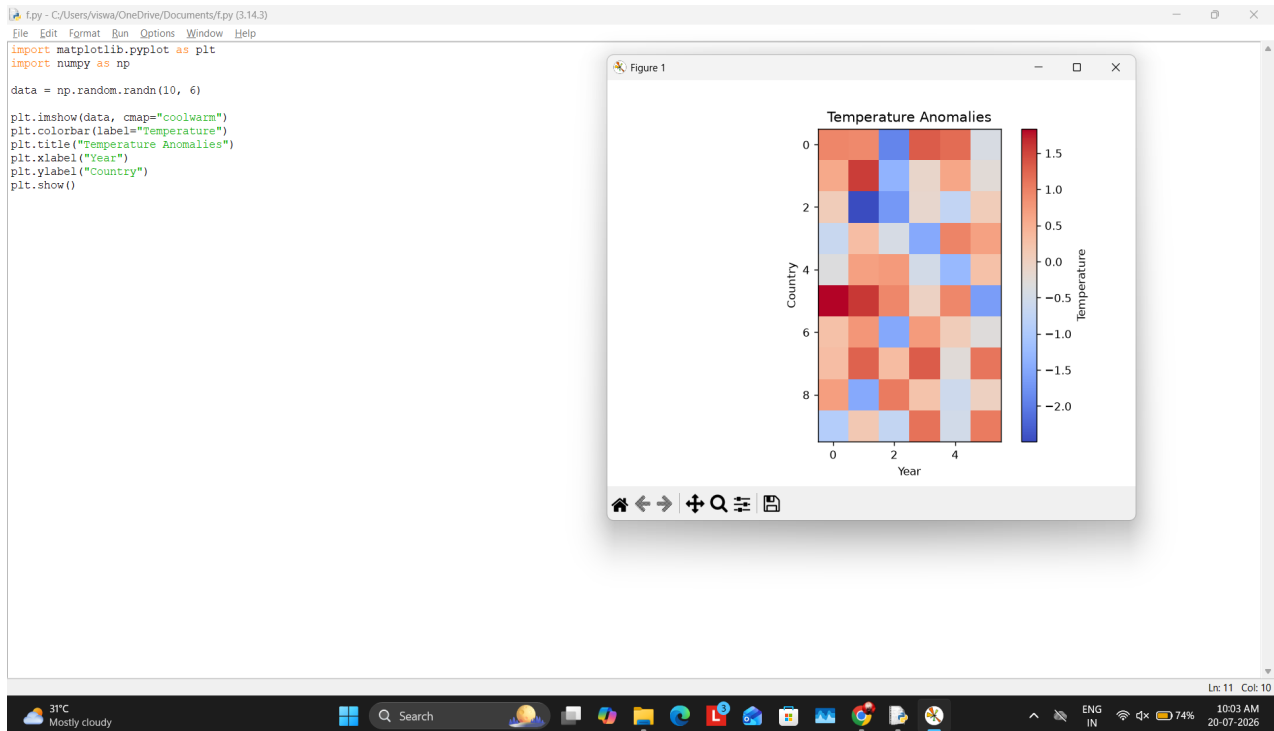
Question 5

```
import matplotlib.pyplot as plt
import numpy as np

data = np.random.randn(10, 6)

plt.imshow(data, cmap="coolwarm")
plt.colorbar(label="Temperature")
plt.title("Temperature Anomalies")
plt.xlabel("Year")
plt.ylabel("Country")
plt.show()
```

Output Screenshot



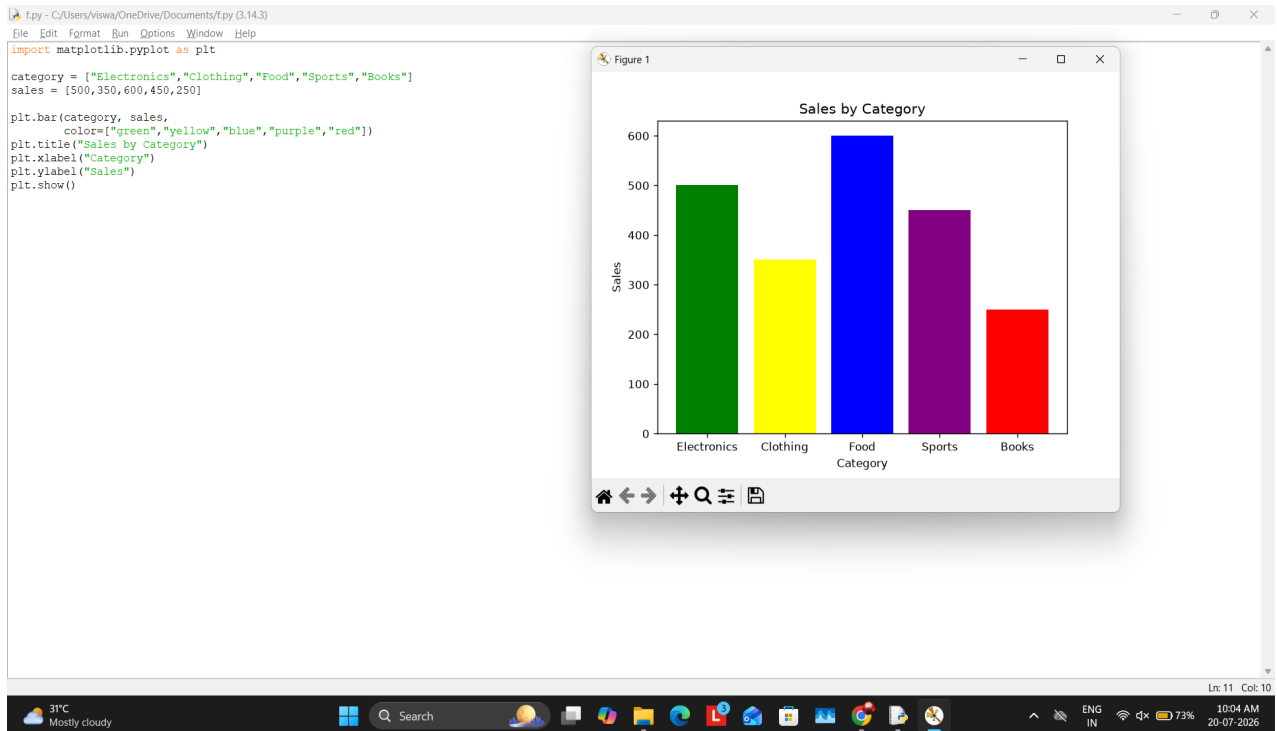
Question 6

```
import matplotlib.pyplot as plt

category = ["Electronics", "Clothing", "Food", "Sports", "Books"]
sales = [500, 350, 600, 450, 250]

plt.bar(category, sales,
         color=["green", "yellow", "blue", "purple", "red"])
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```

Output Screenshot



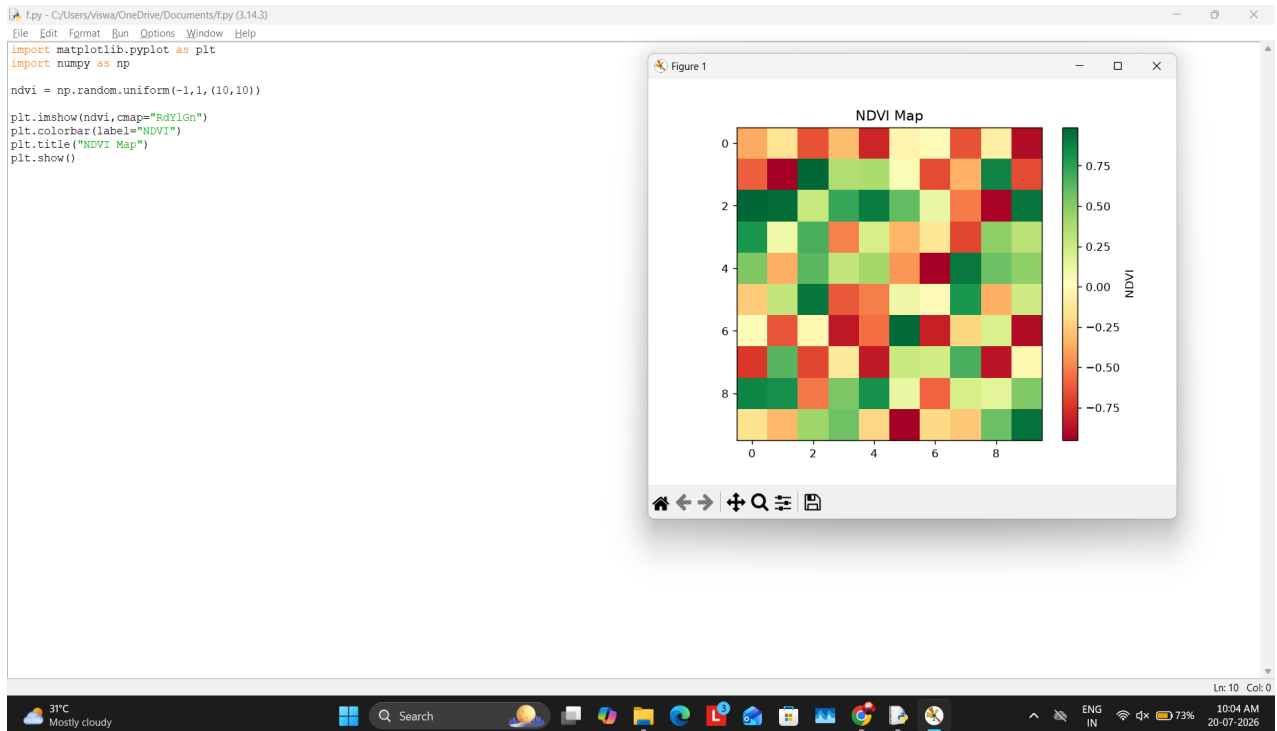
Question 7

```
import matplotlib.pyplot as plt
import numpy as np

ndvi = np.random.uniform(-1,1,(10,10))

plt.imshow(ndvi,cmap="RdYlGn")
plt.colorbar(label="NDVI")
plt.title("NDVI Map")
plt.show()
```

Output Screenshot



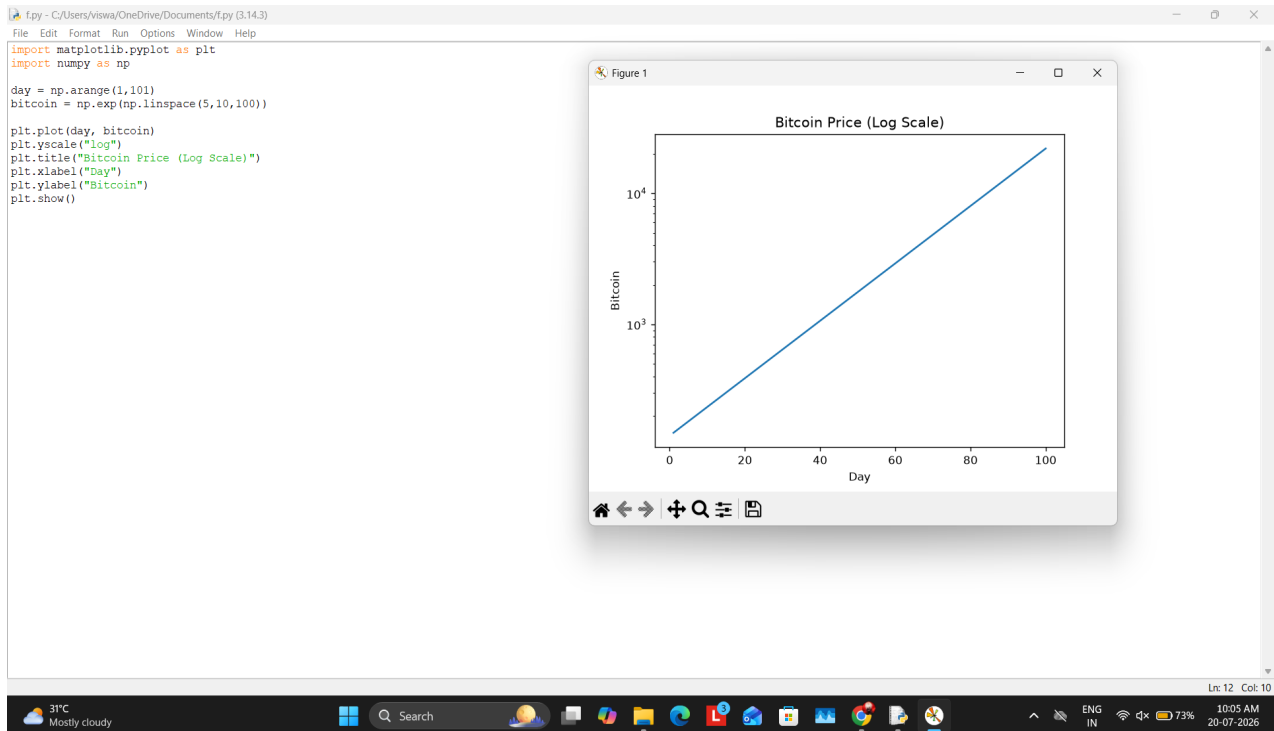
Question 8

```
import matplotlib.pyplot as plt
import numpy as np

day = np.arange(1,101)
bitcoin = np.exp(np.linspace(5,10,100))

plt.plot(day, bitcoin)
plt.yscale("log")
plt.title("Bitcoin Price (Log Scale)")
plt.xlabel("Day")
plt.ylabel("Bitcoin")
plt.show()
```

Output Screenshot



Question 9

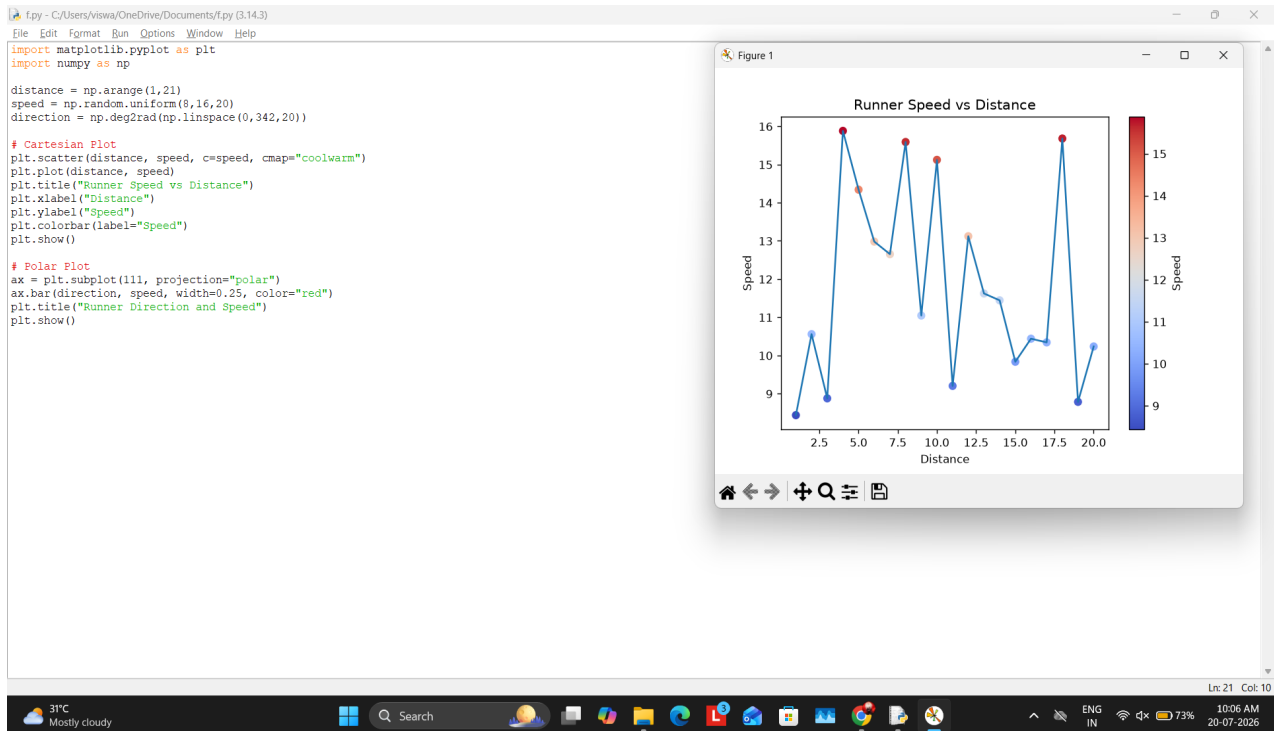
```
import matplotlib.pyplot as plt
import numpy as np

distance = np.arange(1,21)
speed = np.random.uniform(8,16,20)
direction = np.deg2rad(np.linspace(0,342,20))

# Cartesian Plot
plt.scatter(distance, speed, c=speed, cmap="coolwarm")
plt.plot(distance, speed)
plt.title("Runner Speed vs Distance")
plt.xlabel("Distance")
plt.ylabel("Speed")
plt.colorbar(label="Speed")
plt.show()

# Polar Plot
ax = plt.subplot(111, projection="polar")
ax.bar(direction, speed, width=0.25, color="red")
plt.title("Runner Direction and Speed")
plt.show()
```

Output Screenshot



Question 10

```
import matplotlib.pyplot as plt
import numpy as np

district = [f"D{i}" for i in range(1,11)]
rate = np.random.randint(40,100,10)

plt.bar(district, rate, color="limegreen")
plt.title("Vaccination Rate")
plt.xlabel("District")
plt.ylabel("Rate")
plt.show()
```

Output Screenshot

f.py - C:/Users/viwa/OneDrive/Documents/f.py (3.14.3)

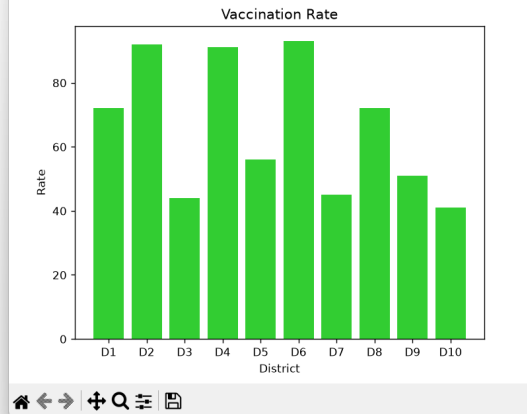
File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt
import numpy as np
```

```
district = [f"D{i}" for i in range(1,11)]
rate = np.random.randint(40,100,10)
```

```
plt.bar(district, rate, color="limegreen")
plt.title("Vaccination Rate")
plt.xlabel("District")
plt.ylabel("Rate")
plt.show()
```

Figure 1



Ln: 11 Col: 10

31°C
Mostly cloudy

Search

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20-07-2026